

EU AI Act Compliance & System Governance

This document explains in simple terms how the IndyRAG system is built to be safe, transparent, and completely compliant with the European Union Artificial Intelligence Act. We have translated complex legal rules into simple, unbreakable system guarantees.

Transparency and Interpretability (Article 13)

Article 13 requires that AI systems be transparent and easy for humans to understand and verify.

- **The Document Registry:**
 - **Purpose:** Keeps a locked, unalterable record of all official instruction manuals uploaded to the system.
 - **Advantage:** Guarantees that no one can secretly alter or tamper with official safety documents. The AI is always reading from the exact, approved materials.
- **Exact Page Citations:**
 - **Purpose:** Forces the AI to point at the exact paragraph, table, or drawing on the physical page where it found its answer.
 - **Advantage:** Eliminates AI "hallucinations" (making up answers). If the AI cannot point to a real spot on a page, it is not allowed to answer. Every claim is 100% verifiable.
- **The Formatting Error Detector:**
 - **Purpose:** Scans and flags any corrupt, garbled, or unreadable pages before the AI reads them.
 - **Advantage:** Prevents the AI from making dangerous mistakes by trying to read corrupted text, keeping the system hygienic and reliable.
- **Operator Reference Footnotes:**
 - **Purpose:** Displays clear footnotes at the bottom of every answer, telling the human operator exactly where each fact came from.
 - **Advantage:** Gives operators instant visibility, allowing them to easily double-check the AI's claims and satisfy transparency rules.

Human Oversight (Article 14)

Article 14 requires that high-risk AI systems remain under the strict control and oversight of human beings.

- **The Layer Ontology (HITL Gate 1):**
- **Purpose:** A predefined, unchangeable master list of functional categories (like electrical or hydraulic systems).
- **Advantage:** Ensures the AI organizes information predictably. It cannot invent new categories or cross boundaries without human permission.
- **The Human Approval Ladder (HITL Gate 3 and Gate 4):**
- **Purpose:** A strict rule stating that no AI-suggested image (Gate 3) or final asset layer (Gate 4) can be used until a human manager explicitly clicks "approve."
- **Advantage:** Places absolute control in human hands, preventing unchecked machine output from entering the final catalog.
- **The Ultimate Integrity Checker:**
- **Purpose:** A final, unskippable electronic gatekeeper that inspects everything before the system publishes the catalog.
- **Advantage:** Guarantees that un-reviewed, unapproved, or tampered data is totally blocked from being exported, securing the system.

Reliability and Task Hygiene (Article 15)

Article 15 requires that AI systems be robust and resilient against errors or crashes.

- **The Agentic Discovery Loops (Discovery Pass 1 & Pass 2):**
- **Purpose:** Secure, temporary staging environments where our visual-discovery and re-cropping agents run multi-turn conversational loops to draft, zoom in, and self-correct bounding boxes.
- **Advantage:** Protects the official catalog from being corrupted during network freezes or crashes, and ensures only high-quality visual crops that pass programmatic checks are ever promoted.
- **Step-by-Step Task Checklists:**
- **Purpose:** Electronic checklists that track the progress of every single task the AI must perform on a page.
- **Advantage:** Prevents the AI from getting distracted, skipping safety steps, or shutting down before it has thoroughly finished its work.
- **Constitutional AI Self-Correction:**

- **Purpose:** Forces the AI to self-evaluate its own images and answers against safety rules while inside the temporary sandbox.
- **Advantage:** Implements privacy and data minimization by design. The system auto-heals its mistakes in memory and deletes unverified, messy drafts instead of permanently storing them on disk.

Logging and Auditability (Article 12)

Article 12 requires systems to keep detailed records of everything they do.

- **The System Flight Recorder:**
- **Purpose:** A continuous, unalterable log file that records every single action, decision, or error the AI makes.
- **Advantage:** Provides a complete, transparent paper trail for safety auditors to inspect exactly how the AI behaved.

The Auditor's Pathway

This section details how a third-party safety inspector or compliance auditor can verify the system's legal alignment in three simple steps:

1. **Check Ingest Integrity:** Run the ultimate integrity checker to mathematically verify that the locked **Document Registry** hashes match the physical blueprint files on the server.
2. **Inspect Human Decisions:** Audit the **Human Approval Ladder** databases to guarantee that every single component published in the catalog carries an explicit, logged signature of human approval.
3. **Verify Claims against Sources:** Read the **Operator Reference Footnotes** generated by the model and traverse the cryptographic hashes and spatial coordinates backwards to verify that the generated claim aligns perfectly with the physical blueprints.

Note: Tracing unstructured execution turns, dropped network connections, or re-cropping loops using the **System Flight Recorder** is separated as a distinct responsibility handled by **Pipeline Developers** to verify code correctness and performance.